DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I

WEEK 13: LECTURE NOTE

ANALYTIC TRIGONOMETRY

Using Fundamental Identities

Proposition 1. ***FUNDAMENTAL TRIGONOMETRIC IDENTITIES***

I. *Reciprocal Identities*

- $\sin\theta = \frac{1}{\csc\theta}$ • $\csc\theta = \frac{1}{\sin\theta}$
- $\cos\theta = \frac{1}{\sec\theta}$ • $\sec\theta = \frac{1}{\cos\theta}$
- $\tan\theta = \frac{1}{\cot\theta}$ • $\cot\theta = \frac{1}{\tan\theta}$

II. *Quotient Identities*

- $\tan\theta = \frac{\sin\theta}{\cos\theta}$
- $\cot\theta = \frac{\cos\theta}{\sin\theta}$

III. *Pythagorean Identities*

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$

IV. *Cofunction Identities*

- $\sin\frac{\pi}{2} - \theta = \cos\theta$ • $\cos\frac{\pi}{2} - \theta = \sin\theta$
- $\tan\frac{\pi}{2} - \theta = \cot\theta$ • $\cot\frac{\pi}{2} - \theta = \tan\theta$
- $\sec\frac{\pi}{2} - \theta = \csc\theta$ • $\csc\frac{\pi}{2} - \theta = \sec\theta$

V. *Even/Odd Identities*

- $\sin(-\theta) = -\sin\theta$ • $\cos(-\theta) = \cos\theta$
- $\tan(-\theta) = -\tan\theta$ • $\cot(-\theta) = -\cot\theta$
- $\sec(-\theta) = \sec\theta$ • $\csc(-\theta) = -\csc\theta$

The above trigonometric identities are easily proved for acute angle θ of a right triangle. It is left as an exercise.

Pythagorean identities are sometimes used in radical form such as

$$\sin\theta = \pm\sqrt{1-\cos^2\theta} \quad \text{or} \quad \tan\theta = \pm\sqrt{\sec^2\theta - 1},$$

where the sign depends on the choice of θ.

One common use of trigonometric identities is to use given values of trigonometric functions to evaluate other trigonometric functions.

Example 2 (Using Identities to Evaluate a Function). *Use the values* $\sec u = -\frac{3}{2}$ *and* $\tan u > 0$ *to find the values of all six trigonometric functions.*

Solution. *Using a reciprocal identity, we have*

$$\cos u = \frac{1}{\sec u} = \frac{1}{-\frac{3}{2}} = -\frac{2}{3}.$$

Using a Pythagorean identity, we have

$$\begin{aligned} \sin^2 u &= 1 - \cos^2 u && \text{Pythagorean identity} \\ &= 1 - \left(-\frac{2}{3}\right)^2 && \text{Substitute} \quad -\frac{2}{3} \quad \text{for} \quad \cos u \\ &= 1 - \frac{4}{9} && \text{Evaluate power} \\ &= \frac{5}{9}. && \text{Simplify} \end{aligned}$$

Because $\sec u < 0$ *and* $\tan u > 0$, *it follows that u lies in Quadrant III. Moreover,* $\sin u$ *is negative when u is in Quadrant III, we can choose the negative root and obtain*

$$\sin u = -\frac{\sqrt{5}}{3}.$$

Now, knowing the values of the sine and cosine, we can find the values of all six trigonometric functions.

- $\sin u = -\dfrac{\sqrt{5}}{3}$,
- $\cos u = -\dfrac{2}{3}$
- $\tan u = \dfrac{\sin u}{\cos u} = \dfrac{\sqrt{5}}{2}$
- $\cot u = \dfrac{1}{\tan u} = \dfrac{2}{\sqrt{5}} = \dfrac{2\sqrt{5}}{5}$
- $\sec u = \dfrac{1}{\cos u} = -\dfrac{3}{2}$
- $\csc u = \dfrac{1}{\sin u} = -\dfrac{3}{\sqrt{5}} = -\dfrac{3\sqrt{5}}{5}$

Example 3 (Simplifying a Trigonometric Expression)**.** *Simplify*

$$\sin x \cos^2 x - \sin x.$$

Solution. *First factor out a common monomial factor and then use a fundamental identity.*

$$\begin{aligned}
\sin x \cos^2 x - \sin x &= \sin x(\cos^2 x - 1) && \text{Factor out monomial factor.}\\
&= -\sin x(1 - \cos^2 x) && \text{Distributive Property}\\
&= -\sin x(\sin^2 x) && \text{Pythagorean identity}\\
&= -\sin^3 x && \text{Multiply.}
\end{aligned}$$

Example 4 (Factoring Trigonometric Expressions)**.** *Factor each expression.*

(a) $\sec^2\theta - 1$ (b) $4\tan^2\theta + \tan\theta - 3$ (c) $\csc^2 x - \cot x - 3$

Solution.

(a) Here the expression is a difference of two squares, which factors as

$$\sec^2\theta - 1 = (\sec\theta - 1)(\sec\theta + 1).$$

(b) This expression has the polynomial form $ax^2 + bx + c$ and it factors as

$$4\tan^2\theta + \tan\theta - 3 = (4\tan\theta - 3)(\tan\theta + 1).$$

(c) Use the identity

$$\csc^2 x = 1 + \cot^2 x$$

to rewrite the expression in terms of the cotangent.

$$\begin{aligned}
\csc^2 x - \cot x - 3 &= (1 + \cot^2 x) - \cot x - 3 && \text{Pythagorean identity}\\
&= \cot^2 x - \cot x - 2 && \text{Combine like terms.}\\
&= (\cot x - 2))(\cot x + 1) && \text{Factor.}
\end{aligned}$$

Example 5 (Simplifying a Trigonometric Expression)**.** *Simplify*

$$\sin x + \cot x \cos x.$$

Solution. *Begin by rewriting in terms of sine and cosine.*

$$\begin{aligned}
\sin x + \cot x \cos x &= \sin x + \left(\frac{\cos x}{\sin x}\right)\cos x && \text{Quotient identity}\\
&= \frac{\sin^2 x + \cos^2 x}{\sin x} && \text{Add fractions.}\\
&= \frac{1}{\sin x} && \text{Pythagorean identity}\\
&= \csc x && \text{Reciprocal identity}
\end{aligned}$$

Verifying Trigonometric Identities

Verifying trigonometric identities is a useful process when you need to convert a trigonometric expression into a form that is more useful algebraically.

Guidelines for Verifying Trigonometric Identities

1. Work with one side of the equation at a time. It is often better to work with the more complicated side first.
2. Look for opportunities to factor an expression, add fractions, square a binomial, or create a monomial denominator.
3. Look for opportunities to use the fundamental identities. Note which functions are in the final expression you want. Sines and cosines pair up well, as do secants and tangents, and cosecants and cotangents.
4. When the preceding guidelines do not help, try converting all terms to sines and cosines.
5. Always try something. Even making an attempt that leads to a dead end provides insight.

Example 6 (Verifying a Trigonometric Identity)**.** *Verify the following identities.*

$$\begin{aligned}
&(a)\quad \frac{\sec^2\theta - 1}{\sec^2\theta} = \sin^2\theta\\
&(b)\quad 2\sec^2\alpha = \frac{1}{1-\sin\alpha} + \frac{1}{1+\sin\alpha}\\
&(c)\quad (\tan^2 x + 1)(\cos^2 x - 1) = -\tan^2 x
\end{aligned}$$

Solution.

(a) Because the left side is more complicated, start with it.

$$\begin{aligned}
\frac{\sec^2\theta - 1}{\sec^2\theta} &= \frac{(\tan^2\theta + 1) - 1}{\sec^2\theta} && \text{Pythagorean identity}\\
&= \frac{\tan^2\theta}{\sec^2\theta} && \text{Simplify.}\\
&= (\tan^2\theta)(\cos^2\theta) && \text{Reciprocal identity}\\
&= (\frac{\sin^2\theta}{\cos^2\theta}(\cos^2\theta) && \text{Quotient identity}\\
&= \sin^2\theta && \text{Simplify}
\end{aligned}$$

(b) The right side is more complicated, so start with it.

$$\begin{aligned}
\frac{1}{1-\sin\alpha} + \frac{1}{1+\sin\alpha} &= \frac{1+\sin\alpha + 1 - \sin\alpha}{(1-\sin\alpha)(1+\sin\alpha)} && \text{Add fractions.}\\
&= \frac{2}{1-\sin^2\alpha} && \text{Simplify.}\\
&= \frac{2}{\cos^2\alpha} && \text{Pythagorean identity}\\
&= 2\sec^2\alpha && \text{Reciprocal identity}
\end{aligned}$$

(c) By applying identities before multiplying, you obtain the following.

$$\begin{aligned}
(\tan^2 x + 1)(\cos^2 x - 1) &= (\sec^2 x)(-\sin^2 x) && \text{Pythagorean identity} \\
&= -\frac{\sin^2 x}{\cos^2 x} && \text{Reciprocal identity} \\
&= -\left(\frac{\sin x}{\cos x}\right)^2 && \text{Property of exponents} \\
&= -\tan^2 x && \text{Quotient identity}
\end{aligned}$$

Solving Trigonometric Equations

To solve a trigonometric equation, use standard algebraic techniques such as collecting like terms and factoring. Our preliminary goal is to isolate the trigonometric function involved in the equation.

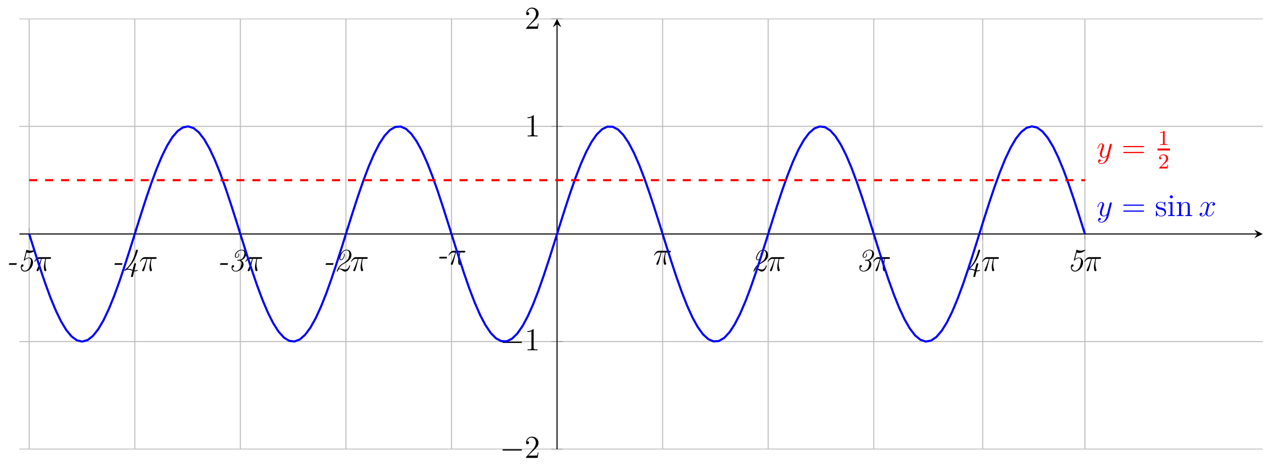
Example 7. *Solve* $2\sin x - 1 = 0$.
Solution.

$$\begin{aligned}
2\sin x - 1 &= 0 && \text{Write original equation.} \\
2\sin x &= 1 && \text{Add 1 to each side.} \\
\sin x &= \frac{1}{2} && \text{Divide each side by 2.}
\end{aligned}$$

To solve for x*, note that the equation has solutions* $x = \frac{\pi}{6}$ *and* $x = \frac{5\pi}{6}$ *in the interval* $[0, 2\pi)$*. Moreover, because* $\sin x$ *has a period of* 2π*, there are infinitely many other solutions, which can be written as*

$$x = \frac{\pi}{6} + 2n\pi \qquad \text{and} \qquad x = \frac{5\pi}{6} + 2n\pi \qquad\qquad \text{General solution}$$

where n *is an integer. This means that any angles that are coterminal with* $\frac{\pi}{6}$ *or* $\frac{5\pi}{6}$ *are also solutions of the equation. Graphically see the intersection points of the graphs of* $y = \sin x$ *and* $y = \frac{1}{2}$ *shown below.*



Example 8 (Factoring an Equation of Quadratic Type)**.** *Find all solutions of*

$$2\sin^2 x - \sin x - 1 = 0$$

in the interval $[0, 2\pi]$.
Solution. *Treating the equation as a quadratic in and factoring produces the following.*

$$\begin{aligned} 2\sin^2 x - \sin x - 1 &= 0 && \text{Write original equation.} \\ (2\sin x + 1)(\sin x - 1) &= 0 && \text{Factor.} \end{aligned}$$

Setting each factor equal to zero, you obtain the following solutions in the interval $[0, 2\pi]$.

$$\begin{aligned} 2\sin x + 1 &= 0 \qquad \text{and} & \sin x - 1 &= 0 \\ \sin x &= -\frac{1}{2} & \sin x &= 1 \\ x &= \frac{7\pi}{6}, \ \frac{11\pi}{6} & x &= \frac{\pi}{2} \end{aligned}$$

Sum and Difference Formulas

Theorem 9 (Sum and Difference Formulas)**.** *Let u and v be radian measures of angles. Then*

$$\begin{aligned} &(a) \quad \sin(u+v) = \sin u \cos v + \cos u \sin v. \\ &(b) \quad \sin(u-v) = \sin u \cos v - \cos u \sin v. \\ &(c) \quad \cos(u+v) = \cos u \cos v - \sin u \sin v. \\ &(d) \quad \cos(u-v) = \cos u \cos v + \sin u \sin v. \\ &(e) \quad \tan(u+v) = \frac{\tan u + \tan v}{1 - \tan u \tan v}. \\ &(f) \quad \tan(u-v) = \frac{\tan u - \tan v}{1 + \tan u \tan v}. \end{aligned}$$

Example 10 (Evaluating a Trigonometric Function)**.** *Find the exact value of*

$$(a) \quad \cos 75^\circ \qquad\qquad (b) \quad \sin\frac{\pi}{12}$$

Solution.

(a) Using the fact that

$$75^\circ = 45^\circ + 30^\circ$$

with the formula for $\cos(u+v)$ *yields*

$$\begin{aligned} \cos 75^\circ &= \cos(45^\circ + 30^\circ) \\ &= \cos 45^\circ \cos 30^\circ - \sin 45^\circ \sin 30^\circ \\ &= \frac{\sqrt{2}}{2}\left(\frac{\sqrt{3}}{2}\right) - \frac{1}{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= \frac{\sqrt{6} - \sqrt{2}}{4} \end{aligned}$$

Try checking this result on your calculator. You will find that $\cos 75^\circ \approx 0.259$.

(b) Using the fact that $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$ *with the formula for* $\sin(u - v)$ *yields*

$$\begin{aligned}
\sin\frac{\pi}{12} &= \sin\left(\frac{\pi}{3} - \frac{\pi}{4}\right) \\
&= (\sin\frac{\pi}{3})(\cos\frac{\pi}{4}) - (\cos\frac{\pi}{3})(\sin\frac{\pi}{4}) \\
&= \frac{\sqrt{3}}{2}\left(\frac{\sqrt{2}}{2}\right) - \frac{1}{2}\left(\frac{\sqrt{2}}{2}\right) \\
&= \frac{\sqrt{6} - \sqrt{2}}{4}
\end{aligned}$$

Example 11. *Find all solutions of*

$$\sin\left(x + \frac{\pi}{4}\right) + \sin\left(x - \frac{\pi}{4}\right) = -1$$

in the interval $[0, 2\pi$.
Solution. *Using sum and difference formulas, rewrite the equation as*

$$\begin{aligned}
\sin x \cos\frac{\pi}{4} + \cos x \sin\frac{\pi}{4} + \sin x \cos\frac{\pi}{4} - \cos x \sin\frac{\pi}{4} &= -1 \\
2\sin x \cos\frac{\pi}{4} &= -1 \\
2\sin x\left(\frac{\sqrt{2}}{2}\right) &= -1 \\
\sin x &= -\frac{1}{\sqrt{2}} \\
\sin x &= -\frac{\sqrt{2}}{2}
\end{aligned}$$

So, the only solutions in the interval $[0, 2\pi]$ *are*

$$x = \frac{5\pi}{4} \qquad \text{and} \qquad x = \frac{7\pi}{4}.$$

Double-Angle Formulas

Theorem 12 (Double-Angle Formulas)**.** *Let* u *be a radian measure of an angle. Then*

$$\begin{aligned}
&(a) \quad \sin(2u) = 2\sin u \cos u. \\
&(b) \quad \cos(2u) = \cos^2 u - \sin^2 u. \\
&(c) \quad \tan(2u) = \frac{2\tan u}{1 - \tan^2 u}.
\end{aligned}$$

Theorem 13 (Power-Reducing Formulas)**.** *Let* u *be a radian measure of an angle. Then*

$$\begin{aligned}
&(a) \quad \sin^2 u = \frac{1 - \cos(2u)}{2}. \\
&(b) \quad \cos^2 u = \frac{1 + \cos(2u)}{2}. \\
&(c) \quad \tan^2 u = \frac{1 - \cos(2u)}{1 + \cos(2u)}.
\end{aligned}$$

Example 14 (Solving a Multiple-Angle Equation)**.** *Solve*

$$2\cos x + \sin 2x = 0.$$

Solution. *Begin by rewriting the equation so that it involves functions of (rather than $2x$). Then factor and solve as usual.*

$2\cos x + \sin 2x = 0$	Write original equation.
$2\cos x + 2\sin x\cos x = 0$	Double-angle formula
$2\cos x(1+\sin x) = 0$	Factor
$2\cos x = 0 \qquad 1+\sin x = 0$	Set factors equal to zero.
$\cos x = 0 \qquad \sin x = -1$	Isolate trigonometric functions.
$x = \frac{\pi}{2}, \frac{3\pi}{2} \qquad x = \frac{3\pi}{2}$	Solutions in $[0, 2\pi]$

So, the general solution is

$$x = \frac{\pi}{2} + 2n\pi \quad \text{and} \quad x = \frac{3\pi}{2} + 2n\pi \qquad \text{General solution}$$

where n is an integer. Try verifying this solution graphically

Example 15 (Reducing a Power)**.** *Rewrite*

$$\sin^4 x$$

as a sum of first powers of the cosines of multiple angles.

Solution.

$\sin^4 x = \left(\sin^2 x\right)^2$	Property of exponents
$= \left(\frac{1-\cos 2x}{2}\right)^2$	Power-reducing formula
$= \frac{1}{4}\left(1 - 2\cos 2x + \cos^2 2x\right)$	Expand binomial.
$= \frac{1}{4}\left(1 - 2\cos 2x + \frac{1+\cos 4x}{2}\right)$	Power-reducing formula
$= \frac{1}{4} - \frac{1}{2}\cos 2x + \frac{1}{8} + \frac{1}{8}\cos 4x$	Distributive Property
$= \frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x$	Simplify.
$= \frac{1}{8}(3 - 4\cos 2x + \cos 4x)$	Factor.

The following theorems are easily proved using the sum and difference formulas discussed.

Theorem 16 (**Half-Angle Formulas**)**.** *Let u be a radian measure of an angle. Then*

$$(a) \quad \sin\frac{u}{2} = \pm\sqrt{\frac{1-\cos u}{2}}$$

$$(b) \quad \cos\frac{u}{2} = \pm\sqrt{\frac{1+\cos u}{2}}$$

$$(c) \quad \tan\frac{u}{2} = \frac{1-\cos u}{\sin u} = \frac{1+\sin u}{\cos u}$$

Theorem 17 (Product-to-Sum Formulas). *Let u and v be radian measures of angles. Then*

$$
\begin{aligned}
(a) \quad \sin u \sin v &= \frac{1}{2}[\cos(u-v) - \cos(u+v)] \\
(b) \quad \cos u \cos v &= \frac{1}{2}[\cos(u-v) + \cos(u+v)] \\
(c) \quad \sin u \cos v &= \frac{1}{2}[\sin(u+v) + \sin(u-v)] \\
(d) \quad \cos u \sin v &= \frac{1}{2}[\sin(u+v) - \sin(u-v)]
\end{aligned}
$$

Example 18 (Solving a Trigonometric Equation)**.** *Find all solutions of*

$$\sin 5x + \sin 3x = 0$$

in the interval $[0, 2\pi)$.
Solution.

$$
\begin{aligned}
\sin 5x + \sin 3x &= 0 && \text{Write original equation.} \\
\sin(4x + x) + \sin(4x - x) &= 0 && \text{Rewrite original equation.} \\
2 \sin 4x \cos x &= 0 && \text{Sum-to-product formula.}
\end{aligned}
$$

By setting the factor $\sin 4x$ *equal to zero, you can find that the solutions in the interval* $[0, 2\pi)$ *are*

$$x = 0,\ \frac{\pi}{4},\ \frac{\pi}{2},\ \frac{3\pi}{4},\ \pi,\ \frac{5\pi}{4},\ \frac{3\pi}{2},\ \frac{7\pi}{4}.$$

The equation $\cos x = 0$ *yields no additional solutions.*